

Characterising Spatial Hallucinations of LLM Advisors in GridWorld Navigation

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Abstract

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Acknowledgements

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Abbreviations

LLM Large Language Model
BFS Breadth-First Search

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Chapter 1

Introduction

1.1 Background

Large language models (LLMs) are increasingly being used as decision-support systems in domains that require reasoning and planning, including robotics, navigation, and embodied agents. In such settings, LLMs are often used to provide step-by-step guidance, such as recommending actions for an agent navigating a physical or simulated environment.

However, despite their strong performance in language and reasoning tasks, LLMs are known to produce incorrect outputs, commonly referred to as hallucinations. In spatial reasoning contexts, these hallucinations manifest as incorrect navigation advice, such as suggesting invalid movements, misinterpreting directions, or failing to account for obstacles. While prior work has demonstrated that these failures occur, there is still limited understanding of the underlying structure of these errors.

1. Introduction

1.2 Problem Statement

Existing research has largely focused on whether LLMs succeed or fail at navigation tasks, typically measured using aggregate metrics such as accuracy or task completion rate. However, these metrics provide limited insight into *how* models fail. In particular, it remains unclear whether spatial errors made by LLMs are random, or whether they follow systematic and predictable patterns.

Understanding the structure of these errors is important, as systematic failure modes could potentially be anticipated and mitigated. Without such analysis, LLM-based systems may remain unreliable in applications requiring spatial reasoning and decision-making.

1.3 Research Objectives

This thesis aims to systematically characterise spatial errors made by LLMs in navigation tasks. To achieve this, the following research questions are investigated:

- Are spatial errors made by LLM advisors random, or do they follow identifiable patterns?
- How do these error patterns change under different task conditions, such as grid size and obstacle density?
- How does prompt framing, specifically egocentric versus allocentric representations, affect model performance?

1.4 Approach Overview

To address these questions, this work adopts a controlled experimental framework based on a GridWorld environment. In this setting, an agent is required to navigate from a start position to a goal position within a two-dimensional grid. At each step, a large language model is prompted to recommend the next action, which is then compared against the optimal action determined using a shortest-path algorithm.

The experiments are conducted under multiple conditions, varying grid size, obstacle density, and prompt framing. For each condition, a large number of navigation episodes are evaluated, and model errors are classified into a structured taxonomy, including categories such as directional confusion and obstacle-related errors.

This design enables fine-grained analysis of model behaviour at the level of individual decisions, rather than relying solely on overall task success.

1.5 Contributions

The main contributions of this thesis are as follows:

- A controlled GridWorld-based framework for evaluating LLM navigation behaviour.
- A method for comparing model-recommended actions against optimal shortest-path actions at each step.
- A fine-grained taxonomy of spatial errors in LLM decision-making.
- An empirical analysis of how error patterns vary across environment configurations and prompt styles.

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1.6 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 reviews related work on spatial reasoning and hallucinations in large language models. Chapter 3 describes the methodology and experimental framework. Chapter 4 outlines the experimental setup and evaluation metrics. Chapter 5 presents the results of the study. Chapter 6 discusses the findings and their implications. Finally, Chapter 7 concludes the thesis and suggests directions for future work.

Chapter 2

Literature Review

2.1 Spatial Reasoning in Large Language Models

2.1.1 General Capabilities of LLMs

Large language models (LLMs) have achieved significant success in natural language processing by learning statistical patterns from large-scale text corpora. Originally developed for linguistic tasks, these models are highly effective at capturing complex relationships and dependencies within sequential data, enabling them to generate coherent and contextually relevant responses.

Recent work has extended the application of LLMs beyond purely textual domains to include tasks that require forms of structured reasoning, including spatial understanding. As highlighted by Zha et al. (2025), the ability of LLMs to model relationships in data has led to growing interest in their use for spatial reasoning tasks, particularly in areas such as robotics, autonomous systems, and virtual environments. In these settings, LLMs are used to interpret natural language de-

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scriptions that involve spatial relationships, object configurations, and navigation instructions.

However, spatial reasoning presents challenges that differ fundamentally from traditional language tasks. Unlike text, spatial environments involve geometric structure, relative positioning, and three-dimensional relationships that are not explicitly encoded in sequential language representations. Zha et al. emphasise that while text can describe spatial arrangements, it often lacks the precision required to fully capture spatial context, particularly in complex or dynamic environments.

To address this limitation, recent research has explored integrating LLMs with additional modalities, such as images, depth information, and point cloud data. These approaches aim to provide models with richer representations of space, enabling them to reason more effectively about spatial structure. The resulting systems demonstrate that LLMs can exhibit a degree of spatial reasoning capability when supported by appropriate representations.

Overall, this body of work suggests that LLMs possess a foundational capacity for reasoning about structured relationships, which can extend to spatial domains. However, this capability remains indirect and dependent on how spatial information is represented and presented, indicating that further work is required to achieve robust and reliable spatial reasoning.

2.1.2 Limitations in Spatial Understanding

Despite recent advances in extending large language models (LLMs) to spatial reasoning tasks, significant limitations remain, particularly in navigation and path planning scenarios. Prior work has shown that LLMs frequently fail even in simple environments, highlighting fundamental weaknesses in their spatial understanding.

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Zhang et al. (2025) demonstrate that LLMs struggle to generate correct navigation strategies even in small grid-based maze environments. Their experiments show that models often fail to reach the goal despite the simplicity of the task, indicating that spatial reasoning remains a challenging problem for LLMs. Notably, these failures occur even when using advanced prompting strategies such as Chain-of-Thought (CoT), Tree-of-Thought (ToT), and ReAct, suggesting that prompting alone is insufficient to overcome these limitations.

A key issue identified in this work is the presence of spatial hallucinations. LLMs frequently generate invalid paths by imagining non-existent connections between locations, such as moving directly through obstacles or assuming unreachable positions are accessible. This reflects a fundamental inability to accurately model spatial relationships based solely on textual descriptions.

In addition to spatial relationship errors, LLMs also exhibit failures related to long-term reasoning. During multi-step planning, models often become trapped in sub-optimal paths, repeatedly visiting the same states or making greedy decisions that prevent them from reaching the goal. These behaviours are attributed to inconsistencies in reasoning over extended sequences of actions, leading to compounding errors over time.

Visual analysis of model behaviour further highlights these limitations. As illustrated in the heatmaps presented by Zhang et al., LLM-based planners tend to cluster around obstacles or repeatedly explore incorrect regions of the environment rather than progressing toward the goal. This suggests that errors are not purely random, but instead follow identifiable patterns in how spatial information is misinterpreted.

Another notable limitation is the tendency of LLMs to favour direct, straight-line paths toward the goal, even when such paths are blocked. This bias leads to system-

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atic errors where the model ignores environmental constraints, further contributing to navigation failure.

Overall, these findings indicate that while LLMs can generate plausible reasoning steps, they lack a robust internal representation of spatial structure. Their errors are systematic and predictable, rather than random, motivating the need for a deeper investigation into the types and patterns of spatial hallucinations exhibited by LLMs.

2.2 Navigation and Planning with LLMs

2.2.1 LLMs as Navigation or Decision Agents

Recent work has explored the use of large language models (LLMs) as decision-making agents in sequential tasks, particularly in embodied and navigation environments. In these settings, LLMs are used to generate high-level action instructions based on textual descriptions of the environment, effectively acting as policy advisors that guide an agent’s behaviour step-by-step.

Zhou et al. (2024) propose the LLM4Teach framework, in which an LLM is used as a teacher agent to provide action guidance for a reinforcement learning (RL) agent. This work demonstrates that LLMs are capable of producing structured action recommendations in response to environmental observations, highlighting their potential for guiding decision-making processes in navigation and planning tasks.

However, despite these capabilities, LLM-based agents exhibit several important limitations. Firstly, LLMs are not inherently specialised for specific tasks or environments. As a result, they often struggle to generate effective action sequences in dynamic or unfamiliar settings. This limitation is partly due to the fact that

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LLMs lack direct interaction with the environment and therefore cannot adapt their behaviour based on real-time feedback.

Secondly, LLM-based decision-making is prone to producing incorrect or suboptimal actions. Zhou et al. show that LLM-only agents frequently generate erroneous decisions when applied to embodied tasks, even when the instructions appear plausible at a surface level. These errors include selecting actions that do not align with task objectives, failing to account for environmental constraints, or producing inconsistent sequences of actions.

The authors further categorise these errors into several types, including incorrect policies, which lead to task failure; inefficient policies, which unnecessarily prolong task completion; and inconsistent policies, which violate logical or contextual constraints. These findings highlight that LLM-generated decisions, while often coherent in language, may lack grounding in the actual environment.

Overall, this line of work demonstrates that LLMs can function as high-level decision-making agents, but their performance remains limited by a lack of task-specific grounding and environmental feedback. This motivates the need for more systematic analysis of the types of errors produced by LLMs when used in navigation and planning contexts.

2.2.2 Path Planning Performance of LLMs

The ability of large language models (LLMs) to perform structured planning tasks has been further examined in the context of pathfinding and navigation. Aghzal et al. investigate whether LLMs can function as effective path planners by directly generating sequences of actions or routes in grid-based environments.

Their findings indicate that while LLMs can sometimes produce valid paths, their

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performance is inconsistent and often unreliable. In simple scenarios, LLMs are capable of generating reasonable plans that reach the goal. However, as task complexity increases—such as with larger grids or the inclusion of obstacles—the quality of the generated paths degrades significantly. This suggests that LLMs struggle to scale their reasoning effectively in structured spatial domains.

A key limitation identified in this work is the tendency of LLMs to produce paths that are suboptimal or incorrect, even when a valid solution exists. Errors include generating unnecessarily long paths, failing to avoid obstacles, or producing sequences that do not reach the intended goal. These behaviours indicate a lack of precise spatial reasoning and an inability to consistently adhere to task constraints.

Importantly, the authors highlight that LLM-generated plans often appear coherent at a linguistic level but fail when evaluated against the underlying environment dynamics. This reinforces the distinction between surface-level reasoning and true task grounding. In other words, an LLM may produce a plausible-looking sequence of directions that does not correspond to a valid or optimal path in the environment.

Overall, this work demonstrates that while LLMs exhibit some capacity for path planning, they do not reliably produce correct or optimal solutions in structured navigation tasks. These findings further motivate the need to systematically analyse the types of spatial reasoning errors made by LLMs, particularly in controlled environments such as GridWorld. —

2.3 Benchmarks for Spatial and Navigation Evaluation

2.3.1 General Spatial Reasoning Benchmarks

A central challenge in studying spatial reasoning in large language models is that strong language performance can mask weaknesses in underlying spatial representation. For this reason, recent work has increasingly emphasised the need for controlled benchmarks that isolate spatial reasoning itself, rather than broader linguistic competence.

Yamada et al. (2024) represent an important step in this direction by proposing a benchmark that evaluates whether LLMs can infer and reason about latent spatial structure from sequential natural-language descriptions. Instead of testing models only on direct factual questions about space, their tasks require the model to track movements through an implied environment and infer the correct spatial outcome. This makes the evaluation more informative, because successful performance depends on maintaining a consistent internal representation of the underlying structure rather than relying on superficial textual cues

An important strength of this framework is that it does not treat spatial reasoning as a single uniform skill. Yamada et al. evaluate models across multiple topologies, including square grids, triangular grids, hexagonal grids, rings, and trees, and show that performance varies substantially across these structures. This is a useful insight for the broader literature: it suggests that claims about “spatial understanding” in LLMs should be made cautiously, because model competence depends heavily on the structure of the environment being reasoned about.

Their results also show that benchmark design itself matters. Performance drops as

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the number of navigation steps increases, indicating that multi-step reasoning places a substantial burden on the model’s ability to maintain spatial consistency over time. In addition, they find that local sequential descriptions can in some cases be easier than presenting the full global map upfront, which suggests that the form in which spatial information is expressed influences measured performance. This is especially relevant for later work, because it implies that apparent model ability may partly reflect representation and prompt design rather than robust spatial competence alone.

Another major contribution of this benchmark is its emphasis on error analysis. Rather than stopping at aggregate accuracy, Yamada et al. examine the structure of model mistakes and show that errors reflect both spatial and non-spatial biases, including sensitivity to spatial proximity and to the temporal order of information in the prompt. This is particularly important for the present thesis, because it demonstrates that analysing how models fail can reveal substantially more than simply reporting whether they succeed.

Overall, Yamada et al. provide a strong general benchmark for probing spatial understanding in LLMs. However, their focus remains on reasoning over implicit spatial structures rather than on action-level navigation decisions in obstacle-aware environments. This leaves open the question of how such reasoning limitations manifest when an LLM must repeatedly choose concrete next-step actions during navigation.

2.3.2 Navigation-Specific Benchmarks

While general spatial reasoning benchmarks evaluate an LLM’s ability to reason about spatial relationships, navigation-specific benchmarks assess whether models can act within an environment through sequential decision-making. Ding et al.

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(2024) introduce MANGO, a benchmark designed to evaluate large language models on navigation and mapping tasks that require continuous interaction with an environment.

In contrast to text-only reasoning tasks, MANGO requires models to generate sequences of actions while maintaining an internal representation of their position and the surrounding environment. This shifts the evaluation from static reasoning to dynamic decision-making, where the model must repeatedly update its state based on previous actions.

A key challenge identified in this work is the difficulty LLMs face in maintaining consistent state representations over time. As tasks progress, models often lose track of their position, leading to incorrect or inconsistent action sequences. This highlights a fundamental limitation in the ability of LLMs to perform sequential reasoning in spatial environments.

The benchmark also demonstrates that LLMs struggle with constructing and maintaining global representations of the environment. In mapping tasks, where the model must infer the layout from partial observations, performance degrades significantly. This suggests that LLMs have difficulty integrating local observations into a coherent global structure.

Similar to findings in general spatial reasoning benchmarks, performance decreases as task complexity increases. Longer navigation sequences lead to greater error accumulation, indicating that multi-step decision-making remains a major challenge for LLMs.

Importantly, the errors observed in MANGO are not purely random. Instead, they exhibit recurring patterns, such as repeated navigation mistakes and inconsistent state updates. These findings suggest that LLM failures in navigation tasks follow identifiable structures, rather than occurring arbitrarily.

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Overall, navigation-specific benchmarks such as MANGO provide a more realistic evaluation of LLM spatial capabilities by requiring action-level decision-making. They highlight the gap between linguistic reasoning and embodied spatial competence, and motivate the need for controlled experimental environments to systematically analyse navigation errors, as explored in this thesis. —

2.4 Spatial Hallucinations in LLMs

2.4.1 Definition of Spatial Hallucination

Hallucination in large language models is commonly understood as the generation of outputs that are plausible in form but inconsistent with the underlying facts or task constraints. In spatial reasoning tasks, this problem takes a more specific form, which may be described as *spatial hallucination*.

In the context of navigation and path planning, spatial hallucination refers to cases where a model produces reasoning or actions that are inconsistent with the true spatial structure of the environment. This includes situations where the model assumes that a path exists when it does not, fails to account for obstacles, misrepresents relative positions, or recommends actions that are incompatible with the actual layout of the space.

Importantly, spatial hallucination is not simply any incorrect answer in a navigation task. Rather, it reflects a mismatch between the model’s inferred representation of the environment and the environment itself. A response may appear coherent at the language level while still being spatially invalid. For example, a model may produce a plausible sequence of directions that would only be correct if the environment were structured differently from the one described.

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For the purposes of this thesis, spatial hallucination is treated as a behavioural phenomenon. That is, it is identified through the model’s observable navigation decisions rather than through assumptions about its internal representations. In a GridWorld setting, spatial hallucination is reflected in action choices that reveal an incorrect understanding of the environment, such as moving toward blocked cells, selecting directions that do not align with the optimal path, or behaving as though obstacles or spatial relationships were different from their true configuration.

This behavioural definition is useful because it allows spatial hallucination to be systematically studied in controlled navigation environments. Rather than asking only whether a model succeeds or fails, it allows analysis of *how* the model’s errors reflect underlying distortions in spatial understanding.

2.4.2 Evidence of Spatial Hallucinations

Empirical evidence for spatial hallucination in large language models is provided by Zhang et al. (2025), who evaluate model performance in grid-based navigation and path planning tasks. Their experiments demonstrate that LLMs frequently produce outputs that are inconsistent with the true spatial structure of the environment, even when the task is relatively simple.

A central finding of this work is that LLMs generate invalid paths that do not correspond to feasible routes in the environment. In many cases, models propose trajectories that pass directly through obstacles or assume connections between locations that do not exist. This behaviour provides clear evidence of spatial hallucination, as the model’s reasoning reflects an incorrect internal representation of the environment.

Importantly, these failures occur even in small and well-defined environments, such

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as 5×5 grid mazes with clearly specified start and goal states. This suggests that spatial hallucination is not merely a result of increasing task complexity, but rather a fundamental limitation in how LLMs represent spatial relationships.

Zhang et al. further show that spatial hallucination manifests in multiple forms. Spatial relationship hallucination occurs when the model misinterprets the layout of the environment, while long-term reasoning hallucination arises when the model fails to maintain consistency across multiple steps. In navigation tasks, this often results in behaviour that appears locally plausible but is globally invalid.

These issues are clearly observable in model behaviour. LLM-based planners frequently become trapped near obstacles, revisit previously explored states, or fail to reach the goal entirely. Notably, such errors persist even when using advanced prompting strategies such as Chain-of-Thought and related reasoning techniques, indicating that prompting alone is insufficient to address the problem.

Overall, this work provides strong empirical support for the existence of spatial hallucinations in LLMs. It shows that model outputs may be linguistically coherent while remaining inconsistent with the true spatial constraints of the environment, reinforcing the need for systematic analysis of navigation errors in controlled settings.

2.4.3 Observed Failure Modes

Beyond demonstrating that large language models exhibit spatial hallucinations, prior work also reveals that these failures follow identifiable and recurring patterns. Rather than producing purely random errors, LLMs tend to make systematic mistakes that reflect underlying limitations in how spatial information is represented and processed.

One common class of errors involves incorrect interpretation of spatial relationships.

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Models may misrepresent the relative positions of locations or objects, leading to incorrect directional decisions. This is closely related to findings from both Zhang et al. (2025) and Yamada et al. (2024), where models generate plausible but spatially inconsistent reasoning.

Another prominent failure mode is the violation of environmental constraints. As shown in navigation and path planning tasks, LLMs frequently generate actions that assume the existence of paths through obstacles or ignore blocked regions entirely. These behaviours indicate a lack of grounded understanding of the environment’s structure, and suggest that models do not reliably incorporate constraints into their reasoning process.

A further source of error arises from difficulties in maintaining consistent state representations over time. In sequential decision-making tasks, models often lose track of their position or revisit previously explored states, resulting in incoherent action sequences. This aligns with observations from navigation benchmarks such as MANGO, where performance degrades as the number of steps increases and errors accumulate over time.

In addition to these spatial and sequential issues, prior work also highlights biases introduced by the structure of the input itself. For example, Yamada et al. show that model predictions can be influenced by the temporal order of information in the prompt, leading to errors that reflect linguistic patterns rather than true spatial reasoning.

Taken together, these findings suggest that LLM spatial errors can be grouped into several broad categories, including relational errors, constraint violations, and temporal inconsistencies. Importantly, these error types are not isolated, but often interact to produce complex failure behaviours in navigation tasks.

This observation motivates the need for a more systematic characterisation of spa-

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tial error patterns at the level of individual actions. By analysing how these failure modes manifest in controlled environments, it becomes possible to move beyond identifying that LLMs fail, toward understanding precisely how and why these failures occur. —

2.5 Directional and Spatial Error Patterns

2.5.1 Cardinal Direction Reasoning

Understanding how large language models reason about directions is a fundamental component of spatial reasoning. Cardinal directions (e.g., north, south, east, west) are central to navigation, map interpretation, and spatial description, making them a useful testbed for evaluating reasoning capabilities in LLMs.

Cohn and Blackwell (2024) provide one of the most focused evaluations of LLM performance on cardinal direction reasoning tasks. Their study distinguishes between simple questions requiring factual recall and more complex scenarios requiring multi-step reasoning. While models achieve high accuracy on simpler tasks, performance degrades significantly as reasoning complexity increases, with the best-performing models achieving only moderate accuracy on structured directional reasoning tasks.

Importantly, their results reveal that LLM errors are not random. Analysis of confusion patterns shows systematic directional errors, particularly confusions between opposite directions such as north and south, and east and west. These findings suggest that LLMs exhibit consistent biases in spatial reasoning rather than purely stochastic behaviour.

The study also identifies specific task conditions that increase error rates. Scenarios involving changes in orientation, such as turning around, or those requiring rea-

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soning about spatial relationships along linear structures (e.g., roads), were found to be particularly challenging. Additionally, performance on inter-cardinal directions (e.g., north-east) was significantly worse than for primary cardinal directions, indicating increased difficulty as spatial reasoning complexity increases.

Another important finding is the influence of prompt formulation. The authors observe that variations in linguistic framing, such as the use of different grammatical persons, can affect model performance, suggesting that LLM spatial reasoning is sensitive to how tasks are described.

Overall, this work demonstrates that while LLMs possess some capacity for spatial reasoning, their performance is inconsistent and highly dependent on task structure. The presence of systematic error patterns and sensitivity to task formulation highlights the need for more fine-grained analysis of LLM spatial behaviour, particularly in decision-making contexts.

2.5.2 Common Error Types in Navigation Tasks

Building on the observed failure modes across prior work, it is possible to identify several recurring categories of errors that characterise how large language models fail in navigation tasks. Rather than occurring randomly, these errors tend to follow consistent patterns that reflect underlying limitations in spatial reasoning and representation.

One common type of error involves incorrect directional decisions. As demonstrated in studies of cardinal direction reasoning, LLMs frequently confuse opposing directions such as north and south or east and west. In navigation settings, these errors manifest as selecting actions that move the agent away from the goal or along an incorrect trajectory, even when the correct direction is locally available.

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A second category of errors relates to violations of environmental constraints. In path planning tasks, models often generate actions that assume the existence of paths through obstacles or fail to recognise blocked regions entirely. This form of error reflects a lack of grounded understanding of the environment, where the model’s inferred spatial structure does not align with the actual layout.

Another prominent failure mode is associated with state tracking and temporal consistency. During multi-step navigation, LLMs frequently lose track of their current position or fail to maintain a coherent representation of previously taken actions. This can lead to behaviours such as revisiting the same locations, entering loops, or producing sequences of actions that are inconsistent over time.

In addition to these issues, LLMs often exhibit suboptimal planning behaviour. Rather than constructing globally optimal paths, models tend to adopt greedy strategies that prioritise immediate progress toward the goal without considering longer-term consequences. This results in inefficient routes, unnecessary detours, or failure to reach the goal altogether when obstacles are present.

Finally, some errors can be attributed to broader representational limitations, where the model forms an incorrect internal understanding of the environment. These errors may manifest as hallucinated spatial relationships, incorrect assumptions about connectivity between locations, or inconsistencies between different parts of the model’s reasoning.

Taken together, these error types highlight that LLM failures in navigation tasks are multifaceted and interconnected. Directional confusion, constraint violations, state tracking failures, and planning errors often occur simultaneously, contributing to complex and persistent failure behaviours. This categorisation provides a foundation for more systematic analysis of navigation errors, and motivates the need to evaluate these patterns in controlled environments such as GridWorld. —

2.6 GridWorld-Based Studies of LLM Spatial Reasoning

2.6.1 LLMs in GridWorld Environments

Recent work has explored the use of large language models (LLMs) in simplified spatial environments such as GridWorld, providing a controlled setting for evaluating navigation behaviour. Martorell (2025) investigates how LLMs perform in a grid-based navigation task and, more importantly, how they internally represent spatial information during decision-making.

In this work, LLMs are evaluated in a 5×5 grid environment where an agent must navigate toward a goal by selecting one of four possible actions at each step. The model receives a textual description of the environment and produces a single action as output, effectively acting as a step-by-step decision-making agent. This setup closely mirrors the use of LLMs as navigation advisors, where decisions are made sequentially based on textual state descriptions.

A key contribution of this study is the systematic comparison of different spatial information representations. The authors evaluate Cartesian representations using explicit coordinates, topographic representations preserving grid structure, and textual descriptions expressed in natural language. Their results show that model performance varies significantly depending on how spatial information is encoded, with Cartesian representations consistently yielding higher success rates and more efficient paths.

Beyond behavioural performance, Martorell focuses on analysing the internal representations of LLMs using probing techniques. The study identifies subsets of model parameters that correlate with spatial features such as the position of the agent and

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the correctness of selected actions. These findings suggest that LLMs develop partial internal representations of spatial structure, with some units encoding abstract spatial information across different contexts.

However, these representations are not fully general. The study shows that spatial representations are strongly influenced by the format in which information is presented, with performance and internal activations varying across representation types. This indicates that LLM spatial reasoning is not entirely invariant, and remains sensitive to prompt design and input structure.

Overall, this work demonstrates that LLMs are capable of performing navigation tasks in GridWorld environments and can develop internal representations of spatial information. However, the focus is primarily on understanding internal model behaviour rather than analysing the types of errors produced during navigation, leaving open questions regarding the structure and consistency of LLM spatial failures in such environments.

2.6.2 Limitations of Existing GridWorld Studies

While prior work has demonstrated that large language models (LLMs) can operate in GridWorld-style environments, existing studies exhibit several important limitations that restrict our understanding of LLM spatial reasoning behaviour.

First, much of the existing work focuses on overall task performance rather than analysing how and why errors occur. For example, studies such as Martorell (2025) primarily evaluate success rates and path efficiency, or investigate internal model representations through probing techniques. While these approaches provide valuable insights into model capabilities and internal structure, they do not offer a detailed characterisation of the types of mistakes made during navigation. As a

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result, it remains unclear whether LLM errors are random or follow consistent and interpretable patterns.

Second, existing GridWorld studies typically operate in highly simplified environments that do not capture key challenges of spatial reasoning. Many tasks involve navigation toward a goal in obstacle-free grids, or define correctness in terms of reducing distance to the target. Although these settings allow for controlled evaluation, they do not require the model to reason about environmental constraints such as blocked paths or alternative routes. Consequently, they do not fully capture the kinds of decision-making errors that arise in more realistic navigation scenarios.

Third, prior work has not systematically examined how different experimental conditions influence LLM behaviour. While some studies explore variations in input representation or prompting format, there is limited analysis of how factors such as environment complexity, obstacle density, or task framing affect error patterns. In particular, the impact of different spatial descriptions—such as egocentric versus allocentric representations—remains underexplored in the context of navigation errors.

Finally, there is a lack of structured frameworks for categorising navigation errors in LLM-based agents. Although individual studies identify specific failure cases, such as incorrect action selection or inconsistent reasoning, these observations are not unified into a coherent taxonomy. This makes it difficult to compare results across studies or to develop a systematic understanding of LLM spatial behaviour.

Taken together, these limitations highlight a key gap in the literature: while existing work demonstrates that LLMs can perform navigation tasks and exhibit identifiable weaknesses, there is no comprehensive analysis of the structure, frequency, and conditions of spatial errors in controlled environments. Addressing this gap requires a systematic, action-level characterisation of LLM errors across varying task condi-

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tions, which forms the central focus of this thesis. —

2.7 Research Gap

The existing literature demonstrates that large language models (LLMs) exhibit both promising capabilities and significant limitations in spatial reasoning tasks. Prior work has shown that LLMs can process structured spatial information and perform navigation-like reasoning under certain conditions. However, these capabilities are inconsistent and highly dependent on task structure, representation, and prompt design.

Research in spatial reasoning benchmarks and navigation environments has consistently highlighted the presence of systematic errors. Studies have shown that LLMs struggle with multi-step reasoning, state tracking, and maintaining consistent representations of spatial environments. In particular, the phenomenon of spatial hallucination—where models generate reasoning or actions that do not correspond to the true environment—has been identified as a key limitation in path planning tasks.

Despite these findings, current work remains limited in its ability to explain how these errors manifest in a structured and generalisable way. Existing studies tend to focus on overall performance metrics, such as success rate or accuracy, or on internal model representations, without providing a detailed analysis of the behavioural patterns underlying model failures. While individual error types have been observed, there is no unified framework for categorising and analysing these errors at the level of individual navigation decisions.

Furthermore, the influence of experimental conditions on spatial error patterns has not been systematically explored. Factors such as environment complexity, obstacle density, and the framing of spatial instructions may significantly affect model

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behaviour, but their impact on error types and frequencies remains unclear. In particular, the distinction between egocentric and allocentric representations has not been thoroughly examined in relation to navigation errors.

As a result, there is a clear gap in the literature: a lack of systematic, action-level characterisation of spatial errors in LLM-based navigation tasks, and an absence of analysis on how these errors vary across different conditions. This limits our ability to understand whether LLM failures are random or follow predictable and structured patterns.

This thesis addresses this gap by introducing a controlled GridWorld environment in which an LLM is used to provide step-by-step navigation decisions. By comparing LLM-generated actions against optimal paths and categorising errors into a structured taxonomy, this work aims to identify recurring patterns in spatial reasoning failures. In addition, the study examines how these error patterns change under varying conditions, including grid size, obstacle density, and prompt framing. Through this approach, the thesis provides a systematic analysis of LLM spatial errors, contributing to a deeper understanding of their behaviour in navigation tasks.

Chapter 3

Methodology

3.1 Overview

This study investigates spatial reasoning errors in large language models (LLMs) through a controlled GridWorld navigation task. The methodology is designed not only to evaluate navigation performance, but to characterise the structure of errors produced by LLMs when acting as sequential decision-making agents.

The experimental framework consists of a discrete two-dimensional grid environment in which an agent must navigate from a start position to a specified goal. Experiments are conducted under two grid sizes (5×5 and 8×8), two obstacle density conditions (10% and 30%), and two prompt formulations (allocentric and egocentric). These controlled variations allow systematic analysis of how spatial complexity and representation influence model behaviour.

At each timestep, the current state of the environment is provided to the LLM in textual form, and the model is required to select a single action corresponding to a movement direction. This interaction is repeated sequentially until the agent reaches

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the goal or a predefined step limit is exceeded.

A key aspect of the methodology is the evaluation of model behaviour at the level of individual actions. Rather than assessing correctness solely based on task completion, each action selected by the LLM is compared against the set of optimal actions at that state.

These optimal actions are computed using a breadth-first search (BFS) algorithm that determines shortest-path distances to the goal. Rather than explicitly enumerating full paths, the method identifies all neighbouring actions that preserve shortest-path optimality. This ensures that multiple equally valid decisions are treated as correct, avoiding bias introduced by selecting a single reference path.

In addition to measuring overall performance metrics such as success rate and step-level accuracy, the methodology focuses on systematic error analysis. Each incorrect action is categorised according to a structured error taxonomy, enabling the identification of recurring patterns in spatial reasoning failures.

The study is conducted under multiple controlled conditions to examine how error patterns vary with environment complexity and representation. In particular, the effect of egocentric versus allocentric spatial descriptions is analysed to determine how different prompt formulations influence model decision-making.

By combining controlled environment design, action-level evaluation, and structured error classification, this methodology provides a framework for analysing whether LLM spatial errors are random or follow consistent and interpretable patterns across different task conditions.

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3.2 Environment

This study employs a custom GridWorld environment designed to provide a controlled and reproducible setting for analysing LLM navigation behaviour. The environment is implemented as a discrete two-dimensional grid.

Experiments are conducted using grid sizes of 5×5 and 8×8 , allowing analysis of how environment scale affects model performance. Obstacle configurations are generated according to predefined densities of 10% and 30%, corresponding to the proportion of grid cells occupied by obstacles. These settings provide controlled variation in both spatial complexity and environmental constraints.

Each environment instance consists of an agent position, a goal position, and a set of obstacle cells. The agent and goal are represented as coordinates (r, c) , where row 0 corresponds to the top of the grid (north) and column 0 corresponds to the left of the grid (west). Rows increase downward (south), and columns increase to the right (east), defining a fixed allocentric coordinate system.

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		Column (c)				
		0	1	2	3	4
Row (r)	0					G
	1		X			
	2			A		
	3					
	4				X	

A = Agent G = Goal X = Obstacle

Figure 1: Example GridWorld environment with a sparse obstacle configuration. The agent (A), goal (G), and obstacle cells (X) are shown in a 5×5 grid. Rows (r) and columns (c) define a fixed allocentric coordinate system.

Obstacles are represented as a binary grid indicating blocked cells that the agent cannot enter. Obstacle placement is generated stochastically based on the specified density, subject to the constraints that the agent and goal positions are not placed on obstacle cells and that a valid path between the agent and goal exists. To ensure reproducibility across experiments, environment instances are initialised using fixed random seeds, allowing identical configurations to be reused across different prompt conditions.

At each timestep, the agent occupies a single grid cell and may attempt to move in one of four directions: north, east, south, or west. These actions correspond to deterministic transitions in the grid, updating the agent’s position by one cell in the chosen direction. If an action would result in movement outside the grid boundaries or into an obstacle cell, the action is treated as invalid, and the agent remains in its current position. Such events are explicitly recorded to enable analysis of constraint violations.

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In addition to its position, the agent maintains an orientation state, represented as one of four directions (north, east, south, west). This orientation is used in egocentric prompt conditions, where actions are expressed relative to the agent’s current facing (e.g., *forward*, *left*). Relative actions are mapped to absolute directions through a deterministic transformation based on the agent’s orientation, ensuring consistency between egocentric and allocentric representations.

The environment operates in a step-based manner, where each action produces a new state along with auxiliary information, including whether the agent has reached the goal, exceeded a maximum step limit, or attempted an invalid move such as colliding with a wall or obstacle. Episodes terminate when the agent reaches the goal or when a predefined maximum number of steps is exceeded.

This controlled environment enables systematic evaluation of LLM navigation behaviour across varying levels of spatial complexity and constraint, forming the foundation for action-level evaluation and subsequent error analysis.

3.3 Model Interaction

The large language model (LLM) is used as a decision-making policy that selects a single navigation action at each timestep based on a textual description of the current environment state. The interaction follows a sequential process, where the model is queried at every step of an episode until termination.

At each timestep, the environment state is converted into a natural language prompt containing the grid size, agent position, goal position, and obstacle locations. Two prompt formulations are used: allocentric and egocentric.

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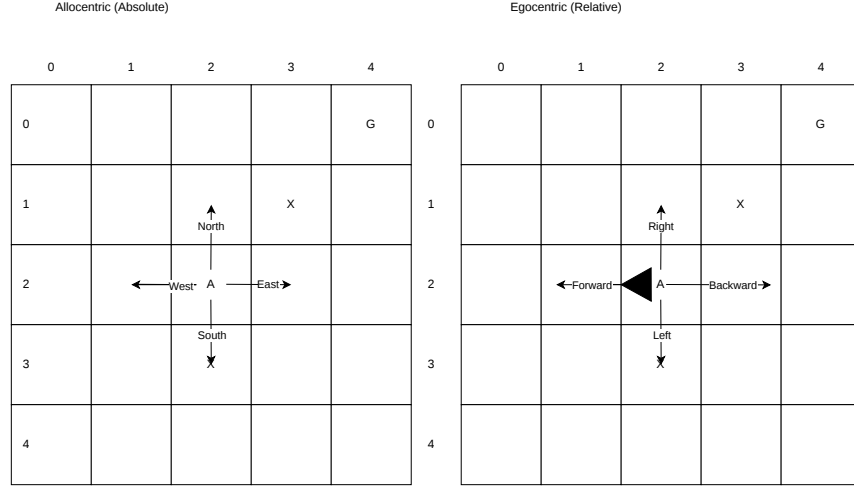


Figure 2: Comparison of allocentric (absolute) and egocentric (relative) spatial representations. In the allocentric condition, actions correspond to fixed global directions, whereas in the egocentric condition, actions are defined relative to the agent's current orientation.

In the allocentric condition, actions are expressed using absolute directions (*north*, *east*, *south*, *west*) within a fixed coordinate system. In the egocentric condition, actions are expressed relative to the agent's current orientation (*forward*, *right*, *back-*

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ward, left), and the agent’s facing direction is explicitly included in the prompt.

The model is instructed to return exactly one action word and no additional explanation. To ensure consistency across runs, the model is queried using deterministic decoding with temperature set to zero, reducing variability in outputs for identical prompts.

Model outputs are processed using a strict parsing procedure. Each response is first normalised by converting to lowercase, removing punctuation, and extracting only the first token. This token is then matched against the set of valid actions for the corresponding prompt condition. Outputs that do not produce a valid action token are classified as `parse_failure` and are not corrected or post-processed. This ensures that evaluation reflects the model’s ability to both follow instructions and produce valid actions.

In the egocentric condition, valid relative actions are deterministically mapped to absolute directions based on the agent’s current orientation. This mapping ensures that all actions are evaluated within a consistent action space, regardless of prompt formulation.

Each model interaction produces a single action, which is executed in the environment. Alongside the action, additional information is recorded, including the raw model output, the parsed action, whether the action is valid, and whether it belongs to the optimal action set. Additional execution details, such as collisions with obstacles or walls, are also logged for subsequent analysis.

This interaction design treats the LLM as a black-box policy operating under strictly defined input and output constraints. By enforcing consistent prompting and deterministic parsing, the methodology ensures that observed errors reflect the model’s spatial reasoning behaviour rather than variability in output formatting or interpretation.

3.4 Optimal Action Computation

To evaluate whether an LLM-selected action is correct, the environment computes the set of optimal next actions from the agent’s current state. This is achieved using breadth-first search (BFS), which determines the shortest valid distance from any grid cell to the goal while respecting obstacle constraints and grid boundaries.

At each timestep, BFS is used to compute the shortest-path distance from the agent’s current position s to the goal g , denoted as $d(s, g)$. The environment then considers all valid neighbouring states s' that can be reached from s by a single action.

An action a leading from s to s' is classified as optimal if it preserves shortest-path optimality. Formally, an action is considered optimal if:

$$1 + d(s', g) = d(s, g)$$

provided that s' lies within the grid and is not an obstacle cell.

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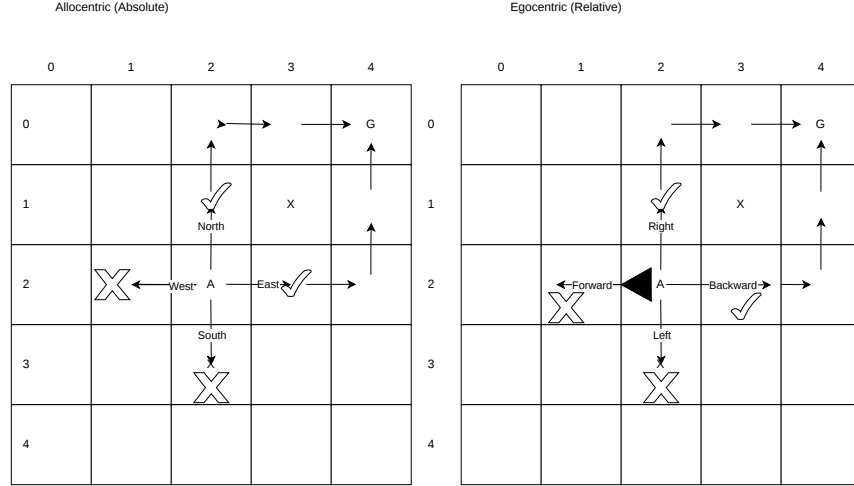


Figure 3: Illustration of optimal and non-optimal actions from a given state. Actions that reduce the shortest-path distance to the goal are considered optimal.

Rather than explicitly enumerating all shortest paths, the implementation identifies all neighbouring actions that satisfy this condition. This allows multiple equally valid actions to be considered correct when more than one shortest path exists between the agent and the goal.

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An LLM action is therefore marked as correct if the parsed action belongs to this optimal action set. This approach provides a fairer and more robust evaluation than comparing against a single reference path, as it avoids penalising valid decisions that lie on alternative shortest routes.

This formulation ensures that correctness is defined at the level of individual decisions, enabling fine-grained analysis of model behaviour and supporting the identification of structured spatial reasoning errors.

3.5 Error Taxonomy

To characterise how large language models (LLMs) fail in navigation tasks, each model decision is classified using a structured error taxonomy. The taxonomy is designed to distinguish between different types of spatial reasoning failures, rather than treating all incorrect actions as equivalent.

At each timestep, the parsed model action is compared against the set of optimal actions computed using the method described in Section 3.4. If the action belongs to this set, it is classified as `correct`. Otherwise, the action is assigned to one of four error categories: `parse_failure`, `obstacle_blindness`, `boundary_error`, or `directional_or_detour_error`.

A `parse_failure` occurs when the model output cannot be interpreted as a valid action. Since only the first token of the model output is used for parsing, any response that does not begin with one of the allowed action words results in a parse failure. These outputs are not corrected or post-processed, ensuring that evaluation reflects the model’s ability to follow instructions and produce valid responses under strict constraints.

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An `obstacle_blindness` error occurs when the model selects an action that would move the agent into an obstacle cell. This indicates a failure to incorporate explicit environmental constraints provided in the prompt.

A `boundary_error` occurs when the selected action would move the agent outside the grid boundaries. This reflects an incorrect understanding of the spatial limits of the environment.

A `directional_or_detour_error` occurs when the selected action is valid in the environment but does not belong to the optimal action set. In this case, the action does not violate any constraints, but it fails to preserve shortest-path optimality. This category captures errors such as moving in the wrong direction, taking unnecessary detours, or selecting locally plausible but globally suboptimal actions.

The error categories are applied hierarchically. Parse failures are identified first, followed by obstacle collisions, boundary violations, and finally valid but non-optimal actions. This ensures that each incorrect decision is assigned a single primary error type.

For aggregate analysis, the implementation adopts a first-error-per-episode approach. Rather than counting every incorrect action within an episode, only the first non-correct decision is recorded. This prevents over-counting caused by repeated failures after the model has already deviated from an optimal path, and provides a clearer indication of the initial failure mode that led to task breakdown.

This taxonomy enables fine-grained analysis of LLM spatial behaviour, allowing errors to be grouped into interpretable categories and supporting investigation into whether failures occur randomly or follow consistent and structured patterns.

Chapter 4

Experimental Setup

4.1 Variables

The experimental design systematically varies three key factors to analyse their impact on LLM navigation performance and error behaviour: grid size, obstacle density, and prompt formulation.

Grid Size. Two grid sizes are used: 5×5 and 8×8 . These settings allow evaluation of how increasing spatial scale affects the model’s ability to perform multi-step reasoning and maintain consistent navigation behaviour.

Obstacle Density. Two obstacle density levels are considered: 10% and 30%. Obstacle density is defined as the proportion of grid cells occupied by obstacles. Higher densities increase environmental constraints, requiring the model to reason about alternative routes rather than relying on direct paths.

Prompt Formulation. Two prompt types are used: allocentric and egocentric. In the allocentric condition, spatial information is presented using absolute coordinates

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and cardinal directions. In the egocentric condition, actions are defined relative to the agent’s current orientation. This variable is used to evaluate how different representations of spatial information influence model decision-making.

These variables are combined to form a set of experimental conditions, enabling systematic comparison of model behaviour across different levels of spatial complexity and representational framing.

4.2 Prompt Conditions

To evaluate how spatial representation affects model behaviour, two prompt formulations are used: allocentric and egocentric. Both prompts describe the same underlying environment state but differ in how spatial information and actions are expressed.

Allocentric Condition. In the allocentric formulation, the environment is described using a fixed coordinate system. The prompt explicitly specifies the grid size, the agent position, the goal position, and the coordinates of all obstacle cells. The coordinate system is defined such that row 0 corresponds to the top of the grid (north) and column 0 corresponds to the left (west), with rows increasing downward and columns increasing to the right. The model is required to select an action from the set $\{north, east, south, west\}$, corresponding to absolute directions in the grid.

Egocentric Condition. In the egocentric formulation, the same environment information is provided, but actions are expressed relative to the agent’s current orientation. In addition to the grid configuration, the prompt includes the agent’s facing direction. The model must select an action from the set $\{forward, right, backward, left\}$, where each action corresponds to movement relative to the current facing direction.

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In both conditions, the prompt includes explicit rules stating that the agent must not move into obstacles or outside the grid, and that the selected action should be part of a shortest valid path to the goal. The model is instructed to return exactly one action word and no additional explanation.

These two prompt formulations allow systematic comparison between absolute and relative spatial representations. This distinction is particularly important for analysing whether LLM spatial reasoning errors are influenced by how directional information is framed, rather than solely by the underlying environment structure.

4.3 Evaluation Metrics

Model behaviour is evaluated using both task-level and error-level metrics. These metrics are designed to measure whether the model completes the navigation task, and to identify the first point at which its behaviour deviates from optimal navigation.

Success Rate. Success rate is defined as the proportion of episodes in which the agent reaches the goal within the maximum allowed number of steps. This provides a high-level measure of task completion under each experimental condition.

Step Correctness. At each timestep, the parsed model action is compared against the optimal action set computed using the method described in Section 3.4. An action is marked as correct if it belongs to this set. This step-level correctness value is logged for every model decision and forms the basis for identifying navigation errors.

First Error per Episode. For aggregate error analysis, the first non-correct event in each episode is recorded. This approach identifies the initial failure mode that

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causes the model to deviate from optimal navigation, rather than over-counting repeated mistakes that may occur after the agent has already gone off course.

Error Type Distribution. First errors are grouped according to the taxonomy described in Section 3.5. The main error categories are `directional_or_detour_error`, `obstacle_blindness`, and `boundary_error`. Parse failures are also recorded when the model output cannot be converted into a valid action.

Episode Length. The number of steps taken in each episode is recorded as `num_steps`. This provides an additional indicator of navigation efficiency, although the primary analysis focuses on success rate and first-error patterns.

Together, these metrics allow the analysis to distinguish between overall task completion and the specific error types responsible for navigation failure.

4.4 Implementation Details

The experimental pipeline is implemented in Python and organised into modular components for environment simulation, prompt construction, model interaction, episode execution, and result analysis.

The LLM policy uses the OpenAI API with the `gpt-4o-mini` model. The model is queried with temperature set to 0.0 to ensure deterministic behaviour, reducing variability in responses to identical prompts. A system instruction is used to enforce that the model returns exactly one allowed action word and no additional explanation.

Each episode follows a sequential interaction loop. The environment is first reset to an initial state, after which a prompt is generated from the current observation. The prompt is passed to the model, and the resulting output is parsed into an

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action. This action is then executed in the environment, producing a new state and associated transition information. This process is repeated until the agent reaches the goal or the maximum step limit is exceeded.

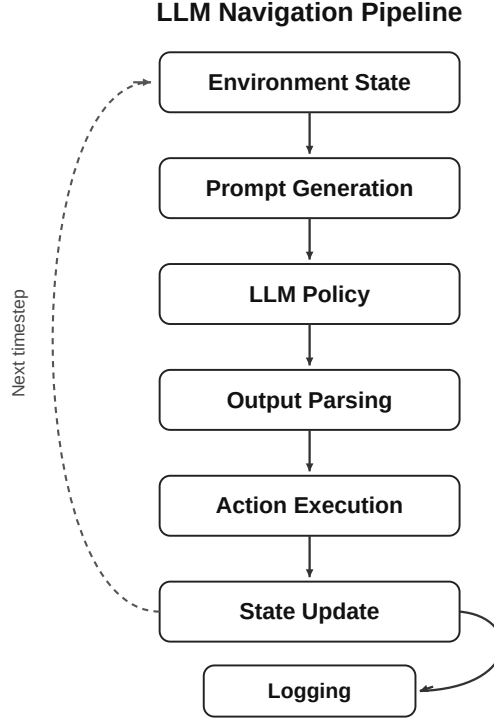


Figure 4: Overview of the LLM navigation pipeline. At each timestep, the environment state is converted into a prompt, processed by the LLM policy, parsed into an action, and executed in the environment to produce a new state. This process repeats sequentially until the episode terminates.

For reproducibility, environment states are generated using fixed random seeds. These states are reused across prompt conditions, ensuring that allocentric and egocentric evaluations are conducted on identical navigation problems. This design enables direct comparison of model behaviour under different spatial representations.

At each timestep, detailed information is recorded, including the agent position, goal position, agent orientation, raw model output, parsed action, optimal action

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set, correctness label, parse failure flag, and collision indicators such as wall or obstacle hits. These logs provide a complete record of model behaviour and form the basis for subsequent analysis.

After all episodes are executed, results are aggregated into structured output files. Separate analysis scripts compute success rates and first-error statistics across experimental conditions. A dedicated plotting pipeline is then used to visualise performance and error distributions, enabling direct comparison across grid sizes, obstacle densities, and prompt formulations.

Chapter 5

Results

5.1 Overall Performance

Figure 5 presents the success rate of the model across all experimental conditions, comparing allocentric and egocentric prompt formulations under varying grid sizes and obstacle densities.

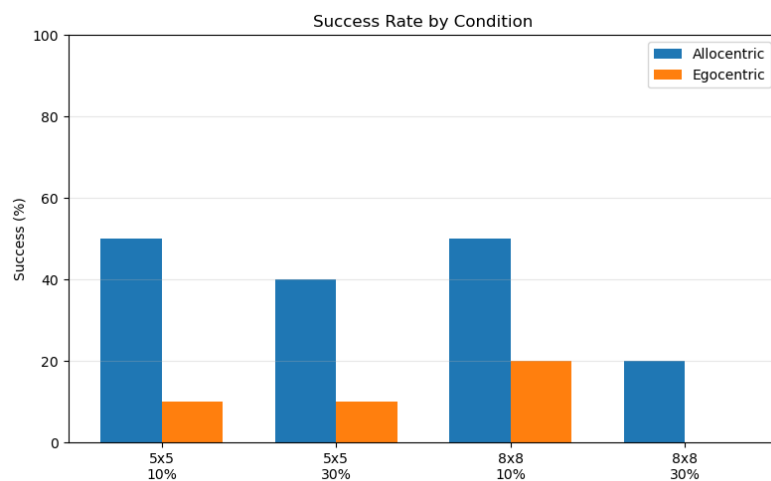


Figure 5: Success rate across grid size, obstacle density, and prompt formulation.

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A clear and consistent performance gap is observed between the two prompt formulations. Under allocentric prompting, the model achieves moderate success rates across most conditions, ranging from 40% to 50% in lower-complexity environments, and decreasing to 20% in the most challenging setting (8×8 grid with 30% obstacle density). In contrast, egocentric prompting results in substantially lower performance, with success rates between 0% and 20%, and complete failure in the most complex condition.

Increasing task difficulty leads to a gradual degradation in performance under allocentric prompting. The transition from lower to higher obstacle density reduces success rates from 50% to 40% in 5×5 grids, and from 50% to 20% in 8×8 grids. This indicates that allocentric reasoning remains relatively robust in simpler environments but becomes increasingly unreliable as environmental constraints increase.

In contrast, egocentric performance is consistently poor even in simpler environments, with success rates of only 10% in both 5×5 conditions. While a slight improvement is observed in the 8×8 , 10% density condition (20%), performance drops to 0% in the most complex setting. This suggests that the model struggles to reliably interpret and apply relative directional instructions, particularly when combined with increased spatial complexity.

The effect of obstacle density appears more pronounced than that of grid size. For example, performance remains stable at 50% when moving from a 5×5 to an 8×8 grid under 10% density, whereas increasing density to 30% results in substantial performance degradation in both cases. This indicates that environmental constraints, rather than spatial scale alone, are the primary driver of task difficulty.

Overall, these results demonstrate that prompt formulation has a dominant effect on navigation performance, with allocentric representations significantly outperforming egocentric ones across all tested conditions.

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5.2 Error Distribution

To better understand the nature of model failures, we analyse the distribution of the first error encountered in each episode across all experimental conditions. Figure 6 shows the proportion of episodes in which each error type occurs, grouped by grid size, obstacle density, and prompt formulation.

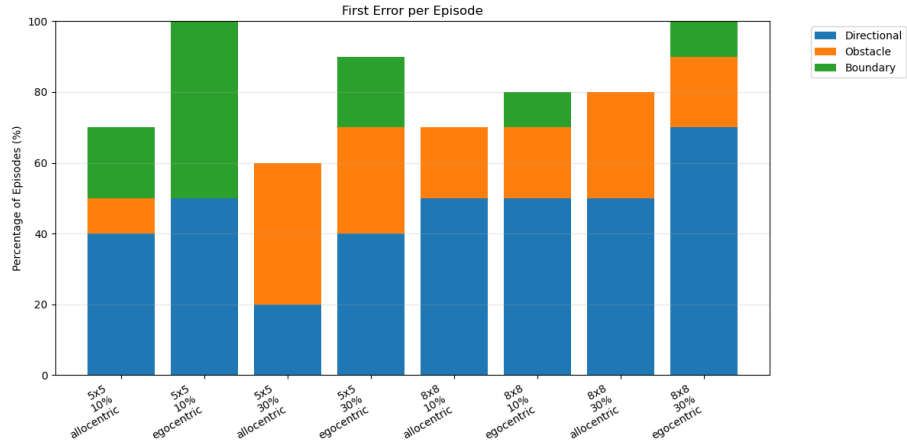


Figure 6: Distribution of first error types across conditions. Errors are categorised as directional errors, obstacle-related errors, and boundary errors.

Across all conditions, directional errors constitute the most frequent failure mode. In most settings, these errors account for a substantial proportion of episodes, often exceeding other error categories. This indicates that the model frequently selects actions that do not align with the shortest-path direction, even when a valid path to the goal exists.

Obstacle-related errors become more prominent as obstacle density increases. In higher-density environments, these errors account for a larger proportion of first failures, reflecting an increased difficulty in incorporating environmental constraints into action selection. This trend is observed across both grid sizes, suggesting that obstacle density is a key factor influencing constraint-related errors.

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Boundary errors occur less frequently overall but exhibit a clear dependence on prompt formulation. These errors are more common under egocentric prompting, particularly in more complex environments. This suggests that representing actions relative to the agent’s orientation introduces additional challenges in maintaining a consistent understanding of spatial limits.

The distribution of error types varies systematically with both environmental complexity and prompt formulation. In simpler conditions, directional errors dominate, reflecting inaccuracies in path-following behaviour. As complexity increases, obstacle-related errors become more prominent, indicating a shift toward failures in constraint handling. Under egocentric prompting, boundary errors further increase, highlighting the additional difficulty introduced by relative spatial representations.

Overall, these results demonstrate that model errors are not random but exhibit structured and condition-dependent patterns. Different types of spatial failures emerge under different conditions, providing evidence that LLM navigation errors can be meaningfully categorised and analysed using a structured taxonomy.

5.3 Prompt Comparison

The results presented in Sections 5.1 and 5.2 demonstrate a consistent and substantial difference in performance between allocentric and egocentric prompt formulations. This section directly compares these two conditions to examine how spatial representation affects both overall success and error behaviour.

Across all experimental settings, allocentric prompting significantly outperforms egocentric prompting. As shown in Figure 5, allocentric success rates remain within the range of 40% to 50% in lower-complexity environments, while egocentric performance remains substantially lower, typically between 0% and 20%. In the most

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challenging condition (8×8 grid with 30% obstacle density), allocentric performance drops to 20%, whereas egocentric performance falls to 0%. This consistent gap indicates that the choice of spatial representation has a dominant effect on navigation performance.

Error distributions further highlight qualitative differences between the two prompt types. Under allocentric prompting, errors are primarily directional, suggesting that the model often misidentifies the correct shortest-path direction despite operating within a fixed coordinate system. In contrast, egocentric prompting exhibits a more diverse range of failure modes. While directional errors remain common, boundary errors are notably more frequent under egocentric conditions, particularly in more complex environments. This suggests that the additional requirement to interpret actions relative to the agent’s orientation introduces further opportunities for failure.

The effect of environmental complexity also differs between the two representations. Allocentric performance degrades gradually as obstacle density increases, indicating a relatively stable but imperfect handling of spatial constraints. Egocentric performance, however, does not exhibit the same gradual degradation pattern. Instead, it remains consistently low even in simpler conditions and collapses entirely in the most complex setting. This suggests that egocentric prompting introduces a fundamental difficulty that is not easily mitigated by reducing task complexity.

Taken together, these results indicate that the performance gap between allocentric and egocentric prompting is not merely quantitative but also qualitative. Allocentric errors are primarily associated with inaccuracies in path selection, whereas egocentric errors additionally reflect difficulties in maintaining a consistent spatial frame of reference. This supports the hypothesis that spatial representation plays a critical role in shaping LLM navigation behaviour, with relative (egocentric) representations leading to more severe and varied forms of spatial failure.

Chapter 6

Discussion

6.1 Interpretation of Findings

6.1.1 Structured Nature of Spatial Errors

The results presented in Chapter 5 provide strong evidence that navigation errors produced by the model are not random but exhibit structured and systematic patterns. The dominance of directional errors across all conditions, combined with the consistent variation of error types under different environmental settings, suggests that model failures arise from identifiable weaknesses in spatial reasoning rather than stochastic variation in output.

Directional errors, which constitute the most frequent failure mode, indicate that the model often selects actions that deviate from the shortest-path direction despite the availability of valid alternatives. This behaviour is observed across both allocentric and egocentric conditions, suggesting a general difficulty in maintaining accurate spatial alignment with the goal during sequential decision-making. At higher obsta-

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cle densities, the increased prevalence of obstacle-related errors further demonstrates that the model struggles to incorporate environmental constraints into its reasoning process.

Beyond aggregate statistics, inspection of individual trajectories provides additional insight into how these errors manifest over time. Consider an example from the egocentric condition in which the agent initially follows a correct path toward the goal but deviates after a single incorrect turn. Following this deviation, the model produces a sequence of suboptimal actions, including repeated reorientations and collisions with environmental boundaries, before eventually recovering and reaching the goal. This pattern illustrates a characteristic failure mode in which a single mistake leads to a cascade of subsequent errors, reflecting difficulty in maintaining a consistent internal representation of spatial state.

Importantly, such behaviour suggests that the model does not reliably correct its course after making an error. Instead, errors tend to propagate across multiple steps, resulting in inefficient or unstable trajectories. This provides further evidence that model failures are structured and arise from limitations in sequential spatial reasoning, rather than isolated incorrect decisions.

Taken together, these findings support the central hypothesis that spatial errors in LLM-based navigation are not random but follow identifiable and interpretable patterns. These structured error modes form the basis for a systematic characterisation of spatial hallucinations in language model-driven agents.

6.1.2 Effect of Spatial Representation

A central finding of this study is the significant impact of spatial representation on navigation performance. Across all experimental conditions, allocentric prompting

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consistently outperforms egocentric prompting, both in terms of overall success rates and the structure of observed errors.

Under allocentric prompting, actions are defined within a fixed global coordinate system. This representation allows each decision to be made directly with respect to the goal location, without requiring the model to maintain additional contextual information. As a result, although directional errors are still common, the model demonstrates relatively stable behaviour, with performance degrading gradually as environmental complexity increases.

In contrast, egocentric prompting introduces an additional layer of complexity by defining actions relative to the agent’s current orientation. In this setting, the model must implicitly track its facing direction across successive steps and correctly transform relative actions into global movements. The results indicate that the model struggles with this requirement, leading to substantially lower success rates and more diverse error patterns.

One key difference is the increased prevalence of boundary errors under egocentric prompting, particularly in more complex environments. These errors suggest that the model fails to consistently maintain an accurate internal representation of its orientation, resulting in actions that violate spatial constraints. This is consistent with the observation that egocentric performance remains low even in simpler conditions, indicating that the difficulty is not solely due to environmental complexity but is inherent to the representation itself.

Furthermore, the degradation in egocentric performance does not follow the same gradual pattern observed in allocentric conditions. Instead, performance remains consistently low across simpler environments and collapses entirely under higher complexity. This suggests that egocentric prompting imposes a fundamental limitation on the model’s ability to perform sequential spatial reasoning, rather than

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simply increasing task difficulty.

Taken together, these findings indicate that spatial representation plays a critical role in shaping LLM navigation behaviour. While allocentric representations allow for relatively stable, goal-directed reasoning, egocentric representations require the maintenance of an internal state that the model does not reliably sustain. This leads to more severe and varied forms of spatial failure, providing strong evidence that representation choice directly influences the emergence of spatial hallucinations.

6.1.3 Impact of Environmental Complexity

The results demonstrate that environmental complexity has a measurable but non-uniform impact on model performance. In particular, obstacle density emerges as a more significant factor than grid size in determining navigation difficulty.

Across both allocentric and egocentric conditions, increasing obstacle density from 10% to 30% results in a clear reduction in success rates. This effect is especially pronounced in larger grids, where performance drops substantially under high-density conditions. The corresponding increase in obstacle-related errors further indicates that the model struggles to consistently incorporate environmental constraints into its action selection process when the environment becomes more cluttered.

In contrast, the effect of grid size alone is comparatively limited. For example, under allocentric prompting at 10% density, performance remains stable when transitioning from a 5×5 to an 8×8 grid. This suggests that simply increasing the spatial extent of the environment does not significantly impair the model’s ability to navigate, provided that the path to the goal remains relatively unconstrained.

These findings indicate that the primary challenge arises not from the scale of the environment, but from the presence of obstacles that restrict available paths. In

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low-density settings, navigation can be achieved through relatively direct movement toward the goal, even in larger grids. However, as obstacle density increases, the model must reason about constrained paths, avoid blocked cells, and maintain consistency over longer sequences of decisions. The results suggest that the model is less effective at handling these constraint-driven requirements.

The interaction between complexity and representation further reinforces this conclusion. While allocentric performance degrades gradually as obstacle density increases, egocentric performance does not exhibit the same pattern, remaining consistently low even in simpler environments. This suggests that the additional representational burden imposed by egocentric prompting outweighs the effects of environmental complexity alone.

Overall, these results highlight that environmental complexity is not a single-dimensional factor. Instead, different aspects of complexity impose distinct challenges on the model. In this setting, constraint-related complexity, as introduced by obstacle density, plays a more critical role than spatial scale in driving navigation errors.

6.1.4 Error Propagation and Recovery

While the analysis in Chapter 5 focuses on the first error encountered in each episode, inspection of full trajectories reveals an additional and important pattern: errors frequently propagate across multiple steps rather than occurring as isolated events.

In many cases, the model initially follows a correct or near-optimal path toward the goal before making a single incorrect decision. Following this deviation, subsequent actions often fail to correct the error, instead producing a sequence of suboptimal or invalid moves. This behaviour is particularly evident in the egocentric condition, where the model must maintain an internal representation of its orientation across

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steps.

For example, in one trajectory from the egocentric setting, the agent begins by moving correctly toward the goal but deviates after an incorrect turn. In the following steps, the model produces a series of actions that result in repeated reorientations and collisions with environmental boundaries, before eventually recovering and reaching the goal. Although the task is ultimately completed, the resulting path is significantly longer and less efficient than the optimal solution. This illustrates how a single error can lead to a cascade of subsequent mistakes, increasing trajectory length and reducing overall reliability.

More severe cases are also observed, in which the model repeatedly selects the same invalid action over multiple steps or enters cyclic behaviour that prevents progress toward the goal. These patterns suggest a failure to update or maintain an accurate internal representation of the environment following an error, leading to persistent misalignment between the model’s actions and the true spatial state.

Importantly, these behaviours are not captured by first-error analysis alone, but they provide critical insight into the model’s sequential decision-making process. The inability to recover from errors indicates that the model lacks robust corrective mechanisms, which would be necessary for stable navigation in dynamic or uncertain environments.

Overall, these observations suggest that spatial errors in LLM-based navigation are not only structured in type but also in their temporal progression. Errors tend to compound over time, leading to increasingly unstable behaviour once the model deviates from an optimal path. This highlights a fundamental limitation in the model’s ability to perform consistent, multi-step spatial reasoning.

6.2 Implications

The findings of this study have several implications for understanding the spatial reasoning capabilities of large language models and their use in sequential decision-making tasks.

First, the structured nature of the observed errors suggests that LLM failures in navigation are not simply the result of random variability in output, but instead reflect systematic limitations in spatial reasoning. The consistent dominance of directional errors, along with predictable shifts in error types under varying environmental conditions, indicates that model behaviour is governed by identifiable patterns. This supports the view that LLM errors can be meaningfully characterised and analysed, rather than treated as unstructured noise.

Second, the strong performance gap between allocentric and egocentric prompting highlights the importance of representation in shaping model behaviour. The difficulty observed in egocentric settings suggests that LLMs struggle to maintain and update internal state information across sequential steps, particularly when reasoning requires transformations between relative and absolute frames of reference. This has broader implications for tasks that depend on consistent internal representations, such as navigation, planning, and interactive decision-making.

Third, the impact of obstacle density demonstrates that LLMs are more sensitive to constraint-driven complexity than to increases in spatial scale. This suggests that models may be capable of handling larger problem spaces when reasoning can remain relatively direct, but struggle when required to integrate multiple constraints and maintain consistency over longer action sequences. Such limitations are likely to affect real-world applications involving structured environments or rule-based decision processes.

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Finally, the observed patterns of error propagation indicate that LLMs lack robust mechanisms for recovery once a mistake has been made. Rather than correcting their behaviour, models often continue to generate suboptimal or invalid actions, leading to compounding failures over time. This has important implications for the deployment of LLMs in sequential or safety-critical settings, where the ability to detect and recover from errors is essential.

Overall, these findings suggest that while LLMs exhibit some capacity for spatial reasoning, their performance is highly sensitive to representation, environmental constraints, and the requirement for consistent multi-step reasoning. Understanding and addressing these limitations will be crucial for improving the reliability of LLM-based agents in navigation and planning tasks.

6.3 Limitations

While this study provides a structured analysis of spatial reasoning errors in large language models, several limitations should be considered when interpreting the results.

First, the experimental environment is based on a simplified GridWorld setting. Although this allows for controlled and systematic evaluation of navigation behaviour, it does not capture the full complexity of real-world spatial tasks. The absence of continuous movement, partial observability, and dynamic environments may limit the generalisability of the findings to more realistic scenarios.

Second, the model is evaluated using a single-step decision framework, in which one action is generated at each timestep without explicit planning or multi-step reasoning. This design isolates the model’s ability to make local decisions but does not assess its capacity for global planning or long-horizon reasoning. As a result,

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the observed errors may reflect limitations in immediate action selection rather than an inability to construct complete navigation strategies.

Third, the analysis focuses on the first error encountered in each episode. While this provides a consistent and interpretable metric for comparing conditions, it does not capture the full trajectory of model behaviour. As discussed in Section 6.1, errors often propagate over multiple steps, and this temporal structure is not fully reflected in the quantitative evaluation.

Fourth, the experiments are conducted using a single model configuration (*gpt-4o-mini*) with deterministic decoding. While this ensures consistency across runs, it limits the ability to generalise findings across different model architectures, sizes, or sampling strategies. Variations in model design or decoding parameters may lead to different error patterns.

Finally, the evaluation assumes access to a complete and accurate description of the environment at each timestep. In more realistic settings, where observations may be partial or noisy, additional challenges would arise that are not captured in this study.

Despite these limitations, the controlled experimental design enables a focused investigation of spatial reasoning behaviour, allowing for clear identification and characterisation of structured error patterns in LLM-based navigation.

Chapter 7

Conclusion

7.1 Summary of Findings

This thesis investigated the spatial reasoning behaviour of large language models in a controlled GridWorld navigation setting, with the aim of characterising the structure of spatial errors produced during sequential decision-making.

The results demonstrate that navigation errors are not random but exhibit consistent and interpretable patterns. Directional errors were found to be the most common failure mode across all conditions, while obstacle-related and boundary errors varied systematically with environmental complexity and prompt formulation. This provides clear evidence that LLM navigation failures can be meaningfully categorised using a structured error taxonomy.

A key finding is that prompt formulation has a significant impact on both performance and error behaviour. Allocentric prompting, which uses a fixed global frame of reference, resulted in substantially higher success rates and more stable behaviour. In contrast, egocentric prompting, which requires reasoning relative to the

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agent’s orientation, led to consistently lower performance and more diverse failure modes. This suggests that LLMs struggle to maintain the internal state required for consistent egocentric reasoning.

Environmental complexity was also found to influence performance, with obstacle density playing a more significant role than grid size. While increasing the size of the environment alone had limited impact under low-density conditions, higher obstacle densities led to substantial increases in constraint-related errors and reduced success rates. This indicates that LLMs are more sensitive to constraint-driven complexity than to spatial scale.

Finally, qualitative analysis of trajectories showed that errors often propagate across multiple steps. Once the model deviates from an optimal path, it frequently fails to recover, resulting in sequences of repeated or compounding errors. This highlights a limitation in the model’s ability to perform stable multi-step reasoning in spatial tasks.

Overall, these findings support the central claim of this thesis: that spatial errors in LLM-based navigation are structured, condition-dependent, and strongly influenced by representation and environmental constraints.

7.2 Contributions

This work makes several contributions to the study of spatial reasoning in large language models.

First, it introduces a controlled experimental framework for evaluating LLM navigation behaviour using a GridWorld environment. This framework enables systematic manipulation of spatial representation, grid size, and obstacle density, allowing for

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detailed analysis of model performance under varying conditions.

Second, the study proposes a structured error taxonomy that categorises navigation failures into directional, obstacle-related, and boundary errors. This taxonomy provides a clear and interpretable way to analyse model behaviour beyond simple success metrics.

Third, the use of a first-error-per-episode evaluation metric ensures that results are not biased by repeated errors within a single episode. This approach provides a more accurate representation of how models fail during navigation and avoids over-counting persistent error behaviours.

Finally, the study demonstrates that prompt framing affects not only overall success rates but also the types of spatial errors produced by the model. This highlights the importance of representation in shaping LLM reasoning and provides new insight into how spatial hallucinations arise in sequential decision-making tasks.

Together, these contributions establish a foundation for systematically characterising spatial reasoning failures in LLM-based agents.

7.3 Future Work

This study represents a controlled pilot investigation of spatial reasoning in LLM-based navigation, and several directions for future work can build on these findings.

One immediate extension is to increase the number of evaluated states. While this study focuses on a limited set of environments for controlled analysis, scaling to a larger number of states would provide more robust statistical validation of the observed patterns.

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Another important direction is to evaluate additional models and configurations. The current study focuses on a single model with deterministic decoding, and extending the analysis to other model architectures, sizes, and sampling strategies would help determine the generality of the observed error patterns.

A further extension is to move beyond GridWorld into more realistic navigation settings. In real-world scenarios, such as indoor robot navigation, agents must operate in environments with dynamic obstacles, partial observability, and sensor noise. For example, a service robot navigating a building would need to avoid blocked corridors, closed doors, and moving people. In this context, directional errors correspond to choosing the wrong path, obstacle-related errors to attempting blocked routes, and boundary errors to invalid movements outside mapped areas.

In such settings, LLM-generated navigation advice should not be executed directly, but instead integrated with traditional planning and sensing systems. GridWorld thus serves as a controlled testbed for identifying structured navigation failures before applying LLMs in more complex and safety-critical environments.

Overall, future work should focus on scaling the evaluation, extending to additional models, and bridging the gap between controlled experimental environments and real-world navigation systems.

Bibliography

Additional Materials

A.1 Prompt Template

A.2 Extra Tables